

Centroids Equations

<u>Type</u>	<u>By Integration</u>	<u>Composite Area / line / Volume</u>
<u>Line Centroid</u>	$\bar{X} = \frac{\int_L \tilde{X} dL}{\int_L dL}$	$\frac{\sum \tilde{X}_i L_i}{L_T}$
<u>Area Centroid</u>	$\bar{X} = \frac{\int_A \tilde{X} dA}{\int_A dA}$	$\frac{\sum \tilde{X}_i A_i}{A_T}$
<u>Volume Centroid</u>	$\bar{X} = \frac{\int_V \tilde{X} dV}{\int_V dV}$	$\frac{\sum \tilde{X}_i V_i}{V_T}$

"P & G" Pappus - Guldinus Equations

$$\text{Surface Area} = A = \sum \theta \bar{r} L$$

$$\text{Volume} = V = \sum \theta \bar{r} A$$

where, θ = angle of rotation of L or A [radians]

$\bar{r}$ = distance from center of rotation to centroid of L or A

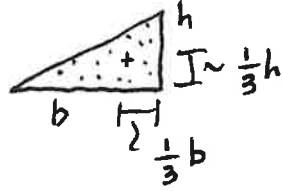
$\theta \bar{r}$ = Arc length A or L centroid moves thru

L = length of line segment

A = Area of Area segment

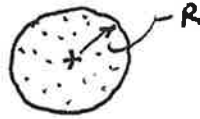
Common Areas and Centroids (And Arcs)

Triangle



$$A = \frac{1}{2}bh$$

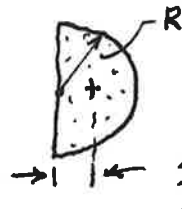
Circular Area



$$A = \pi R^2$$

$$\text{Circumference} = 2\pi R$$

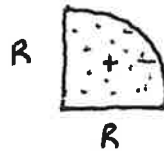
Half Circle



$$A = \frac{1}{2}\pi R^2$$

$$\frac{4R}{3\pi} = \text{Centroid}$$

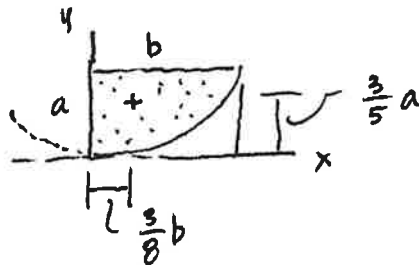
Quarter Circle



$$\frac{4R}{3\pi} = \text{Centroid}$$

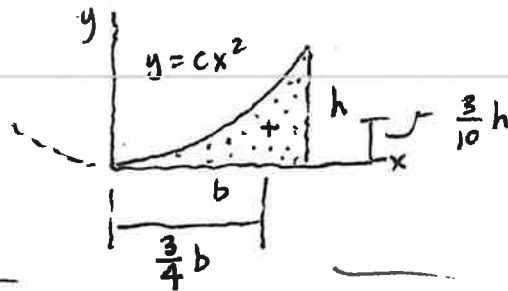
$$A = \frac{1}{4}\pi R^2$$

Parabola (n=2)



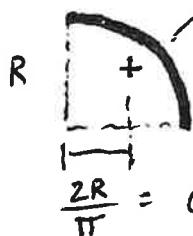
$$A = \frac{2}{3}ab$$

Spandrel (n=2)



$$A = \frac{1}{3}bh$$

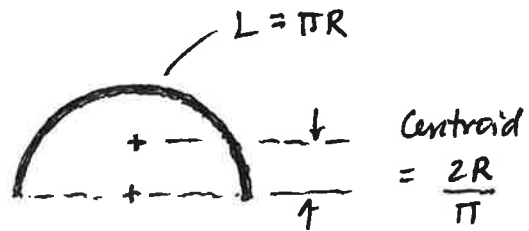
Lines (Arcs)



$$L = \frac{1}{4}2\pi R$$

$$L = \frac{\pi R}{2}$$

$$\frac{2R}{\pi} = \text{Centroid}$$



$$L = \pi R$$

$$\text{Centroid} = \frac{2R}{\pi}$$